A CARTESIAN DIAGRAM OF RINGS STAYS CARTESIAN AFTER LOCALISING AT CORRESPONDING ELEMENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Fibre product of rings and the exactness of the sequence expressing it. used at proof lines 421-426. Provides the starting point of the argument.
- Localisation of a ring at a product of elements and the identification of the localisations appearing in the statement. used at proof lines 426-431. Supplies the equalities B_g = B_h, R'_f = R'_h and R_{s(g)} = R_{t(f)}.
- Exactness of localisation as a functor on modules. the author cites Algebra, Proposition algebra-proposition-localization-exact at proof lines 431-434. Transfers exactness of the sequence to the localised sequence.

Source reference. Original lemma 01Z8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2583–2583. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00CS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 472–472. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Suppose given a cartesian diagram of rings*

$$\begin{array}{ccc} R & \xleftarrow{t} & R' \\ \uparrow{\scriptstyle s} & & \uparrow \\ B & \longleftarrow & B' \end{array}$$

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i.e., $B' = B \times_R R'$. *If* $h \in B'$ *corresponds to* $g \in B$ *and* $f \in R'$ *such that* $s(g) = t(f)$, *then the diagram*

$$\begin{array}{ccc} R_{s(g)} = R_{t(f)} & \xleftarrow[t]{} & (R')_f \\ \uparrow{\scriptstyle s} & & \uparrow \\ B_g & \longleftarrow & (B')_h \end{array}$$

is cartesian too.

Proof. The equality $B' = B \times_R R'$ tells us that

$$0 \to B' \to B \oplus R' \xrightarrow{s,-t} R$$

is an exact sequence of B'-modules. We have $B_g = B_h$, $R'_f = R'_h$, and $R_{s(g)} = R_{t(f)} = R_h$ as B'-modules. By exactness of localization (Algebra, Proposition 00CS) we find that

$$0 \to B'_h \to B_g \oplus R'_f \xrightarrow{s,-t} R_{s(g)} = R_{t(f)}$$

is an exact sequence. This proves the lemma. □